EXT OF MODULE-VALUED SHEAVES COMPUTES PURE EXT OF THE UNDERLYING MODULES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Ext between adequate modules agrees with Ext computed in all O-modules. Same chapter, Lemma lemma-ext-adequate, invoked at proof lines 3014-3017. Replaces the left hand side by an Ext computed in adequate modules.
- The category of adequate functors on Alg_A is equivalent to adequate modules on the affine site. Same chapter, Lemma lemma-adequate-affine, invoked at proof lines 3018-3022. Transfers the computation to adequate functors.
- The injective objects of adequate functors are exactly the pure injective modules. Same chapter, Lemma lemma-pure-injective-injective-adequate, invoked at proof lines 3028-3031. Identifies pure injective resolutions of N with injective resolutions of N-underlined, which gives the result.

Source reference. Original lemma 06ZJ: sources/stacks-adequate/pinned-original-source.tex, source lines 1738–1738. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06ZW: sources/stacks-adequate/pinned-original-source.tex, source lines 2628–2628. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0708: sources/stacks-adequate/pinned-original-source.tex, source lines 2955–2955. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06ZS: sources/stacks-adequate/pinned-original-source.tex, source lines 2474–2474. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Lemma 0.1. *Let $U = \mathrm{Spec}(A)$ be an affine scheme. Let M, N be A-modules. For all i we have a canonical isomorphism*

$$\mathrm{Ext}^i_{Mod(\mathcal{O})}(M^a, N^a) = Pext^i_A(M, N)$$

functorial in M and N.

Proof. Let us construct a canonical arrow from right to left. Namely, if $N \to I^\bullet$ is a pure injective resolution, then $M^a \to (I^\bullet)^a$ is an exact complex of (adequate) $\mathcal{O}$-modules. Hence any element of $\mathrm{Pext}^i_A(M, N)$ gives rise to a map $N^a \to M^a[i]$ in $D(\mathcal{O})$, i.e., an element of the group on the left.

To prove this map is an isomorphism, note that we may replace $\mathrm{Ext}^i_{Mod(\mathcal{O})}(M^a, N^a)$ by $\mathrm{Ext}^i_{Adeq(\mathcal{O})}(M^a, N^a)$, see Lemma 06ZW. Let $\mathcal{A}$ be the category of adequate functors on Alg_A. We have seen that $\mathcal{A}$ is equivalent to $Adeq(\mathcal{O})$, see Lemma 06ZJ; see also the proof of Lemma 06ZS. Hence now it suffices to prove that

$$\mathrm{Ext}^i_{\mathcal{A}}(\underline{M}, \underline{N}) = \mathrm{Pext}^i_A(M, N)$$

However, this is clear from Lemma 0708 as a pure injective resolution $N \to I^\bullet$ exactly corresponds to an injective resolution of $\underline{N}$ in $\mathcal{A}$. □